

X company has gathered the following information.

Units in beginning work in process	0
Units started into production	TK. 40,000
Units in ending work in process	TK. 6,000
Percent complete in ending work in process:	
Conversion costs	40%
Materials	100%
Costs incurred:	
Direct materials	TK. 72,000
Direct labour	TK. 81,000
Overhead	TK. 1,01,000

Instructions:

- (i) Compute equivalent units of production for materials and for conversion costs
- (ii) Determine the unit costs of production.
- (iii) Show the assignment of costs to units transferred out and in process.

Ans:

Given data

- Beginning work in process (WIP): **0 units**
- Units started into production: **40,000 units**
- Ending WIP: **6,000 units**

Completion of ending WIP

- Materials: **100%**
- Conversion costs: **40%**

Costs incurred

- Direct materials: **Tk. 72,000**
- Direct labour: **Tk. 81,000**

- Overhead: **Tk. 1,01,000**

Step 1: Units completed and transferred out

$$\begin{aligned}\text{Units transferred out} &= \text{Units started} - \text{Ending WIP} \\ &= 40,000 - 6,000 = \boxed{34,000 \text{ units}}\end{aligned}$$

(i) Equivalent Units of Production (EUP)

Materials

- Transferred out: $34,000 \times 100\% = 34,000$
- Ending WIP: $6,000 \times 100\% = 6,000$

$$\boxed{\text{EUP (Materials)} = 40,000 \text{ units}}$$

Conversion costs

(Direct labour + Overhead)

- Transferred out: $34,000 \times 100\% = 34,000$
- Ending WIP: $6,000 \times 40\% = 2,400$

$$\boxed{\text{EUP (Conversion)} = 36,400 \text{ units}}$$

(ii) Unit cost of production

Total costs

- Materials = Tk. 72,000
- Conversion costs = Tk. $(81,000 + 1,01,000) = \textbf{Tk. 1,82,000}$

Cost per equivalent unit

Materials

$$\frac{72,000}{40,000} = \text{Tk } 1.80 \text{ per unit}$$

Conversion

$$\frac{1,82,000}{36,400} = Tk. 5.00 \text{ per unit}$$

Total unit cost

$$Tk. (1.80 + 5.00) = Tk. 6.80 \text{ per unit}$$

Not do this in sir slide(sum)

(iii) Assignment of costs

A. Units transferred out (34,000 units)

Cost	Calculation	Tk.
Materials	34,000 × 1.80	61,200
Conversion	34,000 × 5.00	1,70,000
Total		2,31,200

B. Ending work in process (6,000 units)

Cost	Calculation	Tk.
Materials	6,000 × 1.80	10,800
Conversion	2,400 × 5.00	12,000
Total WIP		22,800

Cost reconciliation check

$$\text{Transferred out} + \text{Ending WIP} = 2,31,200 + 22,800 = \boxed{2,54,000}$$

$$\text{Total costs incurred} = 72,000 + 1,82,000 = \boxed{2,54,000}$$

✓ Matched — solution is correct